WEEK 1 · LESSON 1 OF 24 · HANDOUT

What AI Really Is (and Isn't)

Study time: **about 40 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=4RixMPF4xis — open the lesson online to watch inside the course.

🎓 **Go deeper:** Harvard CS50: Artificial Intelligence — full lecture — https://www.youtube.com/watch?v=eey91kzfOZs

Your goal in this lesson: Explain in simple words what artificial intelligence is, tell AI apart from ordinary software, and name three things AI is not.

The one idea to start with

Ordinary software follows rules that a person wrote by hand: *if the amount is above 100,000 RWF, ask for a manager's approval*. Artificial intelligence (AI) is different. Instead of being given every rule, an AI system **learns patterns from examples**.

Think of a spam filter. Years ago, programmers wrote lists of "bad words". Spammers simply changed the words. Today's filters have *learned* from millions of real emails what spam usually looks like — the strange links, the urgent tone, the unknown sender. Nobody wrote those rules. The system found the pattern.

That single idea — **learning patterns from examples instead of following hand-written rules** — explains almost everything you will meet in this course.

Rules vs. learning: a side-by-side view

	Ordinary software	AI system
How it works	Follows rules people wrote	Learns patterns from examples (data)
Example	A calculator, a payroll formula	A spam filter, a voice assistant, ChatGPT
When the world changes	A person must rewrite the rules	It can be retrained on new examples
Can it be wrong?	Only if the rule is wrong	Yes — even when the pattern looked right

Words you will keep hearing

- **Artificial intelligence (AI)** — the broad goal: machines doing tasks that normally need human thinking, like understanding language or recognising images.
- **Machine learning (ML)** — the main method behind today's AI: learning patterns from data.
- **Model** — the "learned pattern" stored inside the computer after training. When you use ChatGPT, you are talking to a model.
- **Data** — the examples a model learns from: text, photos, sounds, numbers.
- **Large language model (LLM)** — a model trained on a huge amount of text so it can read and write language. ChatGPT, Gemini and Claude are built on LLMs.

You already use AI every day

- Your phone keyboard suggests the next word as you type.
- Google Maps predicts how long the trip from Kimironko to Nyabugogo will take.
- Your bank or MoMo app flags a strange transaction.
- YouTube and TikTok decide which video to show you next.
- Your phone camera recognises faces and improves photos at night.

None of these feel like "robots", but all of them learned patterns from data.

What AI is *not*

1. **AI is not magic.** It is mathematics running on computers, using data that people collected.
2. **AI does not "understand" like you do.** A language model produces the most *likely* next words. Usually that is helpful, but it can sound completely confident while being wrong. This is called a **hallucination**.
3. **AI is not conscious and has no goals of its own.** It does what it was built and asked to do.
4. **AI is not always right.** A pattern learned from the past can fail on something new or unusual.

Why this matters for your career

You do not need to *build* AI to benefit from it — just as you did not need to build Excel to use a spreadsheet. The people who get ahead will not all be AI engineers. They will be teachers, accountants, nurses, farmers, students and business owners who know how to **use AI tools well and check their work**. That is exactly what this course teaches.

Remember this: AI learns patterns from examples, it predicts rather than knows, and your job is to use it well and check its work.

Name: _____ Date: _____

Activity

Task 1

Open any free AI assistant (ChatGPT, Gemini, Copilot or Claude).

Task 2

Ask: *"Explain [a topic from your studies or work] to me in simple words, as if I am 15 years old."*

Task 3

Read the answer and find **one statement you should double-check**. Check it using a trusted source (a textbook, an official website or your lecturer).

Task 4

Write two sentences: what the AI did well, and what you had to verify.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What is the key difference between ordinary software and an AI system?

   A. AI learns patterns from examples instead of only following hand-written rules
   B. AI never makes mistakes
   C. Ordinary software cannot use the internet
   D. AI runs on faster computers

2. Which of these is an everyday example of AI?

   A. Your phone keyboard suggesting the next word
   B. A printed timetable
   C. A basic calculator adding 2 + 2
   D. A light switch

3. What is a 'model' in AI?

   A. The computer's screen
   B. A list of rules typed by a programmer
   C. The learned pattern stored after training
   D. A person who poses for photos

4. What is an AI 'hallucination'?

   A. When the screen shows strange colours
   B. When the AI refuses to answer
   C. When the AI becomes conscious
   D. When the AI produces a confident answer that is false

5. Which statement about AI is TRUE?

   A. AI can be useful and still be wrong
   B. AI understands meaning exactly like a person
   C. AI has its own goals and feelings
   D. AI does not need any data

6. According to this lesson, who will benefit most from AI at work?

   A. Nobody — AI will replace everyone
   B. Only large companies
   C. Only AI engineers
   D. People in any field who use AI well and check its work

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